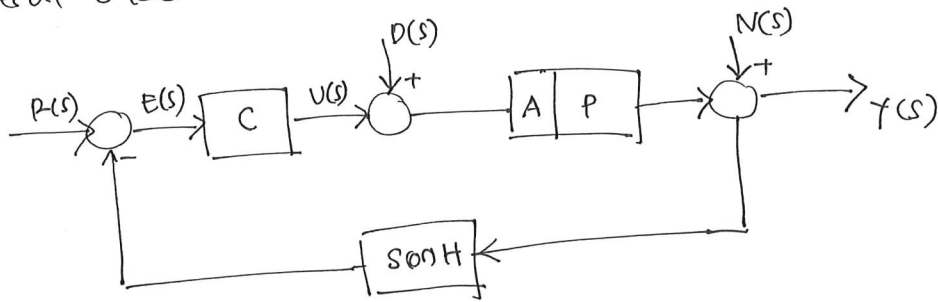


Solution Exercise 01 10004E LP4/25

1. General Block diagram:

①



elements:

$P(s)$: Reference input

$E(s)$: Error (if S or H is unity)

$C(s)$: Controller (feed forward)

$D(s)$: Process Disturbance

$A(s)$: Actuator

$P(s)$: Process or plant

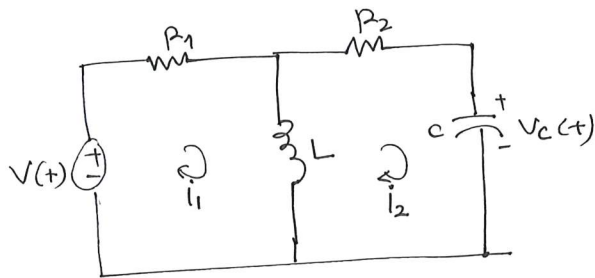
$N(s)$: Measurement noise

$Y(s)$: Output

$S(s)$: sensor / transducer

$H(s)$: feedback controller

2. PLC circuit:



$$P_1 = 1\Omega$$

$$P_2 = 1\Omega$$

$$C = 1F$$

$$L = 1H$$

$V(t)$: input

$V_c(t)$: output
 $i_2(s)$

a. Dynamic equations:

$$(P_1 + Ls) i_1(s) - Ls i_2(s) = V(s) \quad \dots (1)$$

$$-Ls i_1(s) + (P_2 + Ls + \frac{1}{Cs}) i_2(s) = 0 \quad \dots (2)$$

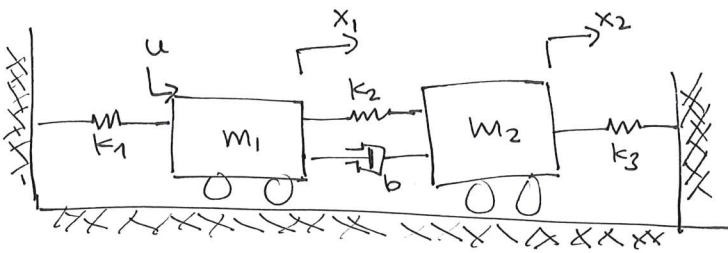
$$V_c(s) = \frac{1}{Cs} i_2(s) \quad \dots (3)$$

$$2b. \frac{I_2(s)}{V} = \frac{s^2}{2s^2 + 2s + 1}$$

$$2c. i_2(t) = \frac{1}{2} e^{-\frac{1}{2}t} (\cos(\frac{1}{2}t) - \sin(\frac{1}{2}t))$$

$$2d. \text{Final value of } \dot{i}_2(t) = 0$$

3. Mechanical system:



$$a. \frac{X_1(s)}{U(s)} = \frac{m_2 s^2 + bs + (k_2 + k_3)}{(m_1 s^2 + bs + (k_1 + k_2))(m_2 s^2 + bs + (k_2 + k_3)) - (bs + k_2)^2}$$

$$b. \frac{X_2(s)}{U(s)} = \frac{\text{Num}(s)}{(m_1 s^2 + bs + (k_1 + k_2))(m_2 s^2 + bs + (k_2 + k_3)) - (bs + k_2)^2}$$

Note: The denominator of both TF is the same.

b. Both $\frac{X_1(s)}{U(s)}$ and $\frac{X_2(s)}{U(s)}$ are 4th order as the characteristic equations are the same.

c. Variables:

states: $x_1, \dot{x}_1, x_2, \dot{x}_2$ Input: u

Parameters

m_1, m_2, k_1, k_2, k_3 and b .

$$4. \quad g(t) = \frac{5}{9} - 5e^{-2t} + \frac{10}{3}te^{-3t} - \frac{40}{9}e^{-3t}$$

5. Steady state value of the output response $y(t)$ for $u(t) = 1$

$$Y(s) = \frac{Y(s)}{U(s)} \cdot U(s)$$

$$= \frac{s+1}{s(s^2+4s+8)} \cdot \frac{1}{s}$$

$$y_{ss}(t) = \lim_{s \rightarrow 0} s Y(s) = \lim_{s \rightarrow 0} s \cdot \frac{s+1}{s^2(s^2+4s+8)} = 1$$

6. $\ddot{y} + 5\dot{y} + 2y^2 = F$

Linearization at $(0,0) \leftarrow$ the origin

$$\ddot{y} + 5\dot{y} = F$$

$$TF: \frac{Y(s)}{F(s)} = \frac{1}{s^2 + 5s} = \frac{1}{s^2 + 5s}$$